

Executive Summary:

What English-Tuned Abliteration Misses in Slovene

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1 Goal

This study asks whether abliteration—removing refusal behaviour by orthogonalising weights against an English-extracted “refusal direction”—transfers to Slovene. Two sibling 12B models from the Gemma-3 family were compared: `google/gemma-3-12b-it` (English-centric) and `cjvt/GaMS3-12B-Instruct` (Slovene continual-pretraining variant), both under NF4 quantisation. The study ran five iterations with pre-registered predictions, independent audits, and 17 falsified hypotheses.

2 What Was Tried

- **Matched Heretic abliteration** (116 TPE trials per model, identical seed) to produce abliterated checkpoints, scored bilingually with keyword markers and LLM judges.
- **Refusal-direction transfer screen:** 2×2 source-language $\times$ evaluation-language ablation matrix, increment tests, and matched-efficacy controls.
- **Surrogacy panels** (exp6, exp7): 200+ Heretic edits per model, measuring whether English refusal outcomes predict Slovene outcomes.
- **Causal mechanism tests** (exp8): activation steering and weight-edit arms to isolate the Slovene residual.
- **Coverage $\times$ strength factorial** (exp9, exp10): 122 and 57 cells testing whether depth coverage or total edit energy governs Slovene refusal survival.
- **Causal write-profile measurement** (exp13, exp14): single-layer edits at all 48 layers, plus an energy-normalised overlap instrument O .
- **Gradient-blind fraction** (exp15): quantifying Heretic’s keyword objective’s structural blindness.
- **Corrected objective** (exp11): replacing keywords with a partial-aware classifier and re-running selection.
- **Cross-model replication** (exp12): Qwen3-8B and Mistral-7B across EN/SL/DE/LT.
- **Three independent audit/recompute passes** (eval1–eval4) checking 400+ numbers.

3 Key Results

1. **GaMS3’s edit works bilingually; Gemma’s does not.** GaMS3’s ablation reduced refusal from 100% to ~6% in both EN and SL. Gemma’s edit reduced EN refusal from 99% to 70% but left SL refusal at 95% (T1, S4 held-out, judged).
2. **The gap is judge-sensitive.** On S5X paired translations (Gemma edit), the EN–SL gap ranges from +0.06 (keyword) to +0.69 (strict judge), with the broad definition at +0.23. The PARTIAL class (54.8% of Gemma edit EN outputs) is the wedge.
3. **Placement orders residual refusal at matched budget.** At matched total removal energy AND matched layer count, where in depth the edit deposits energy orders how much refusal survives—in both languages, in both sibling checkpoints, and in Qwen3-8B. Two cheap one-forward-pass baselines (O_{cos} , O_{band4}) match or beat the expensive causal instrument.
4. **The activation-space depth index fails cross-model.** Spearman = -0.009 over 21 model-language-weight cells. Two baselines beat both the index and the geometric predictor.
5. **The keyword objective is cross-lingually biased.** $\Delta_{\text{lang}} = -1.56$ $[-1.68, -1.44]$: the English keyword rule over-reports in English (FP rate 0.78) and is silent in Slovene (FN rate 0.82). Threshold blindness fraction = 1.0 in both searches.
6. **Utility is unaffected.** All macro utility changes were ≤ 0.20 points; FLORES+ NLL shifts were ≤ 0.002 nats/token.

4 What Failed

Seventeen hypotheses were falsified (F1–F17). The most consequential:

- **Depth coverage explains the gap (F1):** $\Delta R^2 = 0.040$, LOO -0.002 .
- **Broad-and-weak beats narrow-and-strong (F2):** falsified opposite (-0.10).
- **Corrected objective produces a better edit (F5):** dose-scaling the keyword edit achieved comparable gap reduction at lower KL.
- **r_prior direction carries Slovene residual (F7):** causal cut $0.036 < \text{best random control } 0.074$.
- **Count-based depth index predicts cross-model (F3):** Spearman -0.009 .
- **Anchor search is blinder overall (F17):** paired GBF difference $+0.006$; judge-referenced reverses.

5 Best Result

The firm positive is a **matched-budget placement contrast**: at fixed total edit energy and fixed layer count, depositing the edit in band 13–24 (the causal write-profile peak) reduces Slovene refusal from 0.92 to 0.27, while the same energy in band 33–48 leaves it at 0.92. This holds for Gemma, GaMS3, and Qwen3-8B. At the refusal floor of shipped edits, however, dose—not placement—is the active variable ($p = 0.774$). Two cheap baselines match the expensive instrument, so the practical

recommendation is: use energy-weighted cosine or band-mass overlap to select layer sets, not the full causal write profile.

On the guard side, the official RefusEU ASR (Llama-Guard-3-8B + PolyGuard agreement) for the Gemma edit is EN 0.738 vs SL 0.103 = +0.635 gap, confirming the behavioural asymmetry with an independent safety metric.

6 Open Questions

1. **Native-speaker validation.** All Slovene material is machine-translated (Gemini-2.5-flash); five packets (640 items) await native-speaker labelling.
2. **Slovene judge certification.** The SL judge gate was not met ($\kappa = 0.723 < 0.80$); all SL gap claims are JUDGE_SENSITIVE.
3. **NF4 confound.** Paired bf16 vs NF4 gap: bf16 +0.60 vs NF4 +0.45 ($n = 20$, not resolved). The asymmetry is at least as large in bf16.
4. **Two checkpoints $\neq$ a population.** Nothing is attributed to continual pretraining, instruction tuning, or any training stage.
5. **Optimizer variance.** One seed per model; prompt-level CIs do not capture run-to-run optimizer variance.
6. **Placement at the floor.** The placement contrast is strongest mid-range; at shipped-edit refusal levels, dose dominates ($p = 0.774$ for placement swap).
7. **Defect rate.** 136 numbers independently re-derived; defect rate 15.4% [10.3%, 22.5%] on previously unaudited sections. No sign reversals.